

Independent $\text{Spin}(10)$ Invariant-Basis and PQ Audit

Action correction, global quotient, and the domain-wall gate

Route-F four-dimensional mainline

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Abstract

We audit the scalar action of $3(16_F) + 54_{\mathbb{R}} + 126_{\mathbb{C}} + 10_{\mathbb{C}} + 1_{\mathbb{C}}$ without assuming that the previously printed tensor list is complete. The representation ring and an exhaustive PQ-neutral multidegree enumeration find one missing renormalizable invariant, $\chi_7 \Phi_{ij} \phi_i \phi_j S^* + \text{h.c.}$. Thus P54PQ-v1 is superseded by P54PQ-v2; the parameter counts become 60/49/48 and three scalar CP phases survive. We also obtain $\mathcal{A}_{\text{Spin}(10)^2-\text{PQ}} = -6$, the QCD instanton coefficient $\hat{N} = -12$, and physical $N_{\text{DW}} = 3$ after the diagonal $\mathbb{Z}_4$ center identification. The latter is a genuine cosmological debt, not a convention-dependent factor of two.

1 Audit contract

Write

$$A = 54, \quad B = 126, \quad C = \overline{126}, \quad H = 10, \quad \bar{H} = 10^*, \quad S = 1, \quad \bar{S} = 1^*, \quad (1)$$

with PQ charges

$$q(A, B, C, H, \bar{H}, S, \bar{S}) = (0, 2, -2, -2, 2, -4, 4). \quad (2)$$

Because a bosonic monomial uses symmetric powers whenever identical fields occur, a tensor-product singlet count without the appropriate permutation projection is not sufficient. We therefore use both the full products and their symmetric parts.

The required product data are [1]

$$\begin{aligned} 10 \otimes 10 &= 1_s \oplus 45_a \oplus 54_s, \\ 54 \otimes 54 &= 1 \oplus 45 \oplus 54 \oplus 660 \oplus 770 \oplus 1386, \\ 126 \otimes 10 &= 210 \oplus 1050, \\ 126 \otimes 54 &= 126 \oplus 1728 \oplus 4950, \\ 126 \otimes 126 &= 54 \oplus 945 \oplus 1050 \oplus 2772 \oplus 4125 \oplus 6930, \\ 126 \otimes \overline{126} &= 1 \oplus 45 \oplus 210 \oplus 770 \oplus 5940 \oplus 8910. \end{aligned} \quad (3)$$

The dimensions determine the permutation projections:

$$\begin{aligned} \text{Sym}^2(54) &= 1 \oplus 54 \oplus 660 \oplus 770, & \wedge^2(54) &= 45 \oplus 1386, \\ \text{Sym}^2(126) &= 54 \oplus 1050 \oplus 2772 \oplus 4125, & \wedge^2(126) &= 945 \oplus 6930. \end{aligned} \quad (5)$$

For example, $54 + 1050 + 2772 + 4125 = 8001 = 126 \cdot 127/2$.

2 Complete invariant basis through dimension four

The following table gives one representative of a complex-conjugate pair; “multiplicity” is the dimension of the singlet space at that multidegree. The coefficient labels coincide with the corrected action card.

Degree	multidegree	multiplicity	coefficient(s)
2	A^2	1	μ^2
2	BC	1	ν^2
2	$H\bar{H}$	1	ξ_0^2
2	$S\bar{S}$	1	μ_s^2
3	A^3	1	c
3	$AH\bar{H}$	1	ξ_3
3	$H^2\bar{S} + \text{h.c.}$	1	χ_6 complex
4	A^4	2	a, b
4	A^2BC	2	α, β
4	$A^2H\bar{H}$	2	η_0, η_2
4	$A^2S\bar{S}$	1	χ_3
4	B^2C^2	4	$\lambda_0, \lambda_2, \lambda_4, \lambda'_4$
4	$BCH\bar{H}$	2	γ_1, γ_2
4	$BCS\bar{S}$	1	χ_2
4	$H^2\bar{H}^2$	2	ξ_1, ξ_2
4	$H\bar{H}S\bar{S}$	1	χ_5
4	$S^2\bar{S}^2$	1	χ_1
4	$AB^2S + \text{h.c.}$	1	χ_4 complex
4	$AH^2\bar{S} + \text{h.c.}$	1	χ_7 complex
4	$B^2CH + \text{h.c.}$	1	η_1 complex
4	$B^2H^2 + \text{h.c.}$	1	η_3 complex

The nontrivial multiplicities follow by common-channel intersection:

$$\begin{aligned}
A^2BC &: (1 \oplus 54 \oplus 660 \oplus 770) \cap (1 \oplus 45 \oplus 210 \oplus 770 \oplus \dots) = 1 \oplus 770, \\
A^2H\bar{H} &: \text{Sym}^2(54) \cap (10 \otimes 10) = 1 \oplus 54, \\
B^2C^2 &: \text{Sym}^2(126) \cap \text{Sym}^2(\overline{126}) = 54 \oplus 1050 \oplus 2772 \oplus 4125, \\
BCH\bar{H} &: (126 \otimes \overline{126}) \cap (10 \otimes 10) = 1 \oplus 45, \\
AB^2S &: 54, \quad AH^2\bar{S} : 54, \quad B^2CH : 1050, \quad B^2H^2 : 54.
\end{aligned} \tag{6}$$

For A^4 , the orthogonal conjugation invariants of a real traceless symmetric matrix at degree four are $(\text{tr } A^2)^2$ and $\text{tr } A^4$; the four channels in $\text{Sym}^2(54)$ do not give four independent fully symmetric fourth powers.

The new term is therefore unavoidable:

$$\boxed{V \supset \chi_7 \Phi_{ij} \phi_i \phi_j S^* + \chi_7^* \Phi_{ij} \phi_i^* \phi_j^* S.} \tag{7}$$

Equation (3) gives the unique 54 contraction, while $0 + 2(-2) + 4 = 0$ verifies PQ neutrality. Its multidegree differs from every other operator, so equations of motion or integration by parts cannot remove it from a non-derivative renormalizable basis.

For completeness, the machine ledger enumerates all $6 + 12 + 26 = 44$ PQ-neutral multidegrees of degrees two, three, and four. Every entry not in the table fails a displayed channel intersection. Two useful examples are

$$BH : \quad 126 \otimes 10 = 210 \oplus 1050 \not\cong 1, \quad ABC : \quad 126 \otimes \overline{126} \not\cong 54. \tag{8}$$

The second statement is independently checked using random self-dual five-forms: the symmetric-traceless part of $K_{ij} = \Sigma_{iabcd} \Sigma_{jabcd}^*/4!$ vanishes at machine precision, whereas its antisymmetric 45 part is nonzero.

3 Corrected parameter quotient

There are 24 real scalar coefficients and five complex ones, hence 34 raw scalar reals. Under independent rephasings of (Σ, ϕ, S) the five complex-coupling phase vectors are

$$v_{\eta_1} = (1, 1, 0), \quad v_{\eta_3} = (2, 2, 0), \quad v_{\chi_4} = (2, 0, 1), \quad v_{\chi_6} = v_{\chi_7} = (0, 2, -1). \quad (9)$$

Their exact rank is two and their common null direction is the PQ action. Thus three invariant phases remain; a convenient basis is

$$\begin{aligned} \delta_1 &= \arg \eta_3 - 2 \arg \eta_1, \\ \delta_2 &= \arg \chi_4 + \arg \chi_6 - 2 \arg \eta_1, \\ \delta_3 &= \arg \chi_7 - \arg \chi_6. \end{aligned} \quad (10)$$

Including two gauge/topological reals and 24 raw Yukawa reals gives

$$\boxed{N_{\text{raw}} = 60, \quad N_{\text{classical}} = 49, \quad N_{\text{quantum PQ}} = 48.} \quad (11)$$

The second count removes the two scalar rephasings and the nine-dimensional $U(3)$ family orbit. The third trades the unified theta angle against the anomalous PQ origin.

4 PQ anomaly from first principles

For left-handed Weyl fermions define

$$\mathcal{A}_{G^2-\text{PQ}} \delta^{AB} = \sum_f q_f \text{tr}_{R_f} (T^A T^B). \quad (12)$$

With $\text{tr}_{16}(T^A T^B) = 2\delta^{AB}$ and three $q_{16} = -1$ families,

$$\boxed{\mathcal{A}_{10} \equiv \mathcal{A}_{\text{Spin}(10)^2-\text{PQ}} = 3(-1)T(16) = -6.} \quad (13)$$

The QCD result can be checked without unification. One family contains two colored Weyl triplets in q_L and one antitriplet in each of u^c, d^c . Using $T(3) = T(\bar{3}) = 1/2$,

$$\mathcal{A}_3^{(1)} = (-1)[2T(3) + T(\bar{3}) + T(\bar{3})] = -2, \quad \mathcal{A}_3 = -6. \quad (14)$$

For an integer instanton number ν , a PQ rotation by α changes the measure by

$$\exp(i\hat{N}\alpha\nu), \quad \hat{N} = 2\mathcal{A}_3 = -12. \quad (15)$$

The gravitational and cubic global-anomaly ledger is similarly

$$\mathcal{A}_{\text{grav}^2-\text{PQ}} = 3 \cdot 16(-1) = -48, \quad \mathcal{A}_{\text{PQ}^3} = 3 \cdot 16(-1)^3 = -48. \quad (16)$$

Scalars do not contribute to chiral triangle anomalies.

5 Global form and domain-wall number

The charge normalization alone gives $\gcd(|q_B|, |q_H|, |q_S|) = 2$ and the often quoted naive result $|\hat{N}|/2 = 6$. This is not yet the number of physically distinct vacua because the matter spinor fixes the gauge group to $\text{Spin}(10)$, whose center is $\mathbb{Z}_4$.

Choose its generator z to act on $(16, 54, 126, 10, 1)$ as

$$z = (i, 1, -1, -1, 1). \quad (17)$$

At $\alpha = \pi/2$, the PQ action is

$$e^{i\alpha q} = (-i, 1, -1, -1, 1). \quad (18)$$

Their product is the identity on every field. Hence the faithful global group is

$$\frac{\text{Spin}(10) \times U(1)_{\text{PQ}}}{\mathbb{Z}_4^{\text{diag}}}, \quad (19)$$

and the physical PQ orbit has $\alpha \sim \alpha + \pi/2$. The QCD potential is proportional to $\cos(12\alpha + \bar{\theta})$, so its number of minima in one physical orbit is

$$N_{\text{DW}} = \frac{12(\pi/2)}{2\pi} = 3. \quad (20)$$

This agrees with the center-quotient prescription and explicit axion alignment analysis of Ref. [2]. It also explains why dividing the naive answer by four would be wrong: the naive six has already divided by the common scalar charge two. Starting from $|\hat{N}| = 12$, the single diagonal $\mathbb{Z}_4$ quotient gives three directly.

$N_{\text{DW}} = 3 > 1$ means that a post-inflation PQ transition is excluded without an additional repair. Viable options are: PQ breaking before or during inflation with no later restoration; extra colored chiral matter chosen to change the anomaly while preserving unification; or an explicitly declared small bias whose induced strong-CP shift is separately bounded. None is part of P54PQ-v2; therefore the domain-wall problem remains an honest model-level debt.

6 Gate decision

The independent invariant and anomaly audits are closed, but their result is not “the old action is confirmed.” Instead:

1. P54PQ-v1 is superseded because it omitted (7).
2. P54PQ-v2 is the only action on which P1 may construct a stationary point, Hessian, or spectrum.
3. The exact gauge-orbit calculation gives rank 24 for the 54 direction and rank 33 for $54 + 126$, leaving the 12 Standard-Model generators. The P1 Hessian must therefore contain exactly 33 gauge Goldstones before gauge fixing, plus the PQ axion direction prior to QCD effects.
4. $N_{\text{DW}} = 3$ is carried forward as a phenomenological blocker, not silently reset to one.

References

- [1] R. Slansky, “Group theory for unified model building,” [Phys. Rept. 79 \(1981\) 1–128](#).
- [2] A. Ernst, A. Ringwald, and C. Tamarit, “Axion Predictions in $SO(10) \times U(1)_{\text{PQ}}$ Models,” [JHEP 02 \(2018\) 103](#).
- [3] K. S. Babu and S. Khan, “A Minimal Non-Supersymmetric $SO(10)$ Model,” [arXiv:1507.06712](#).